

Rudin, *Principles of Mathematical Analysis*

Chapter 8 — Some Special Functions

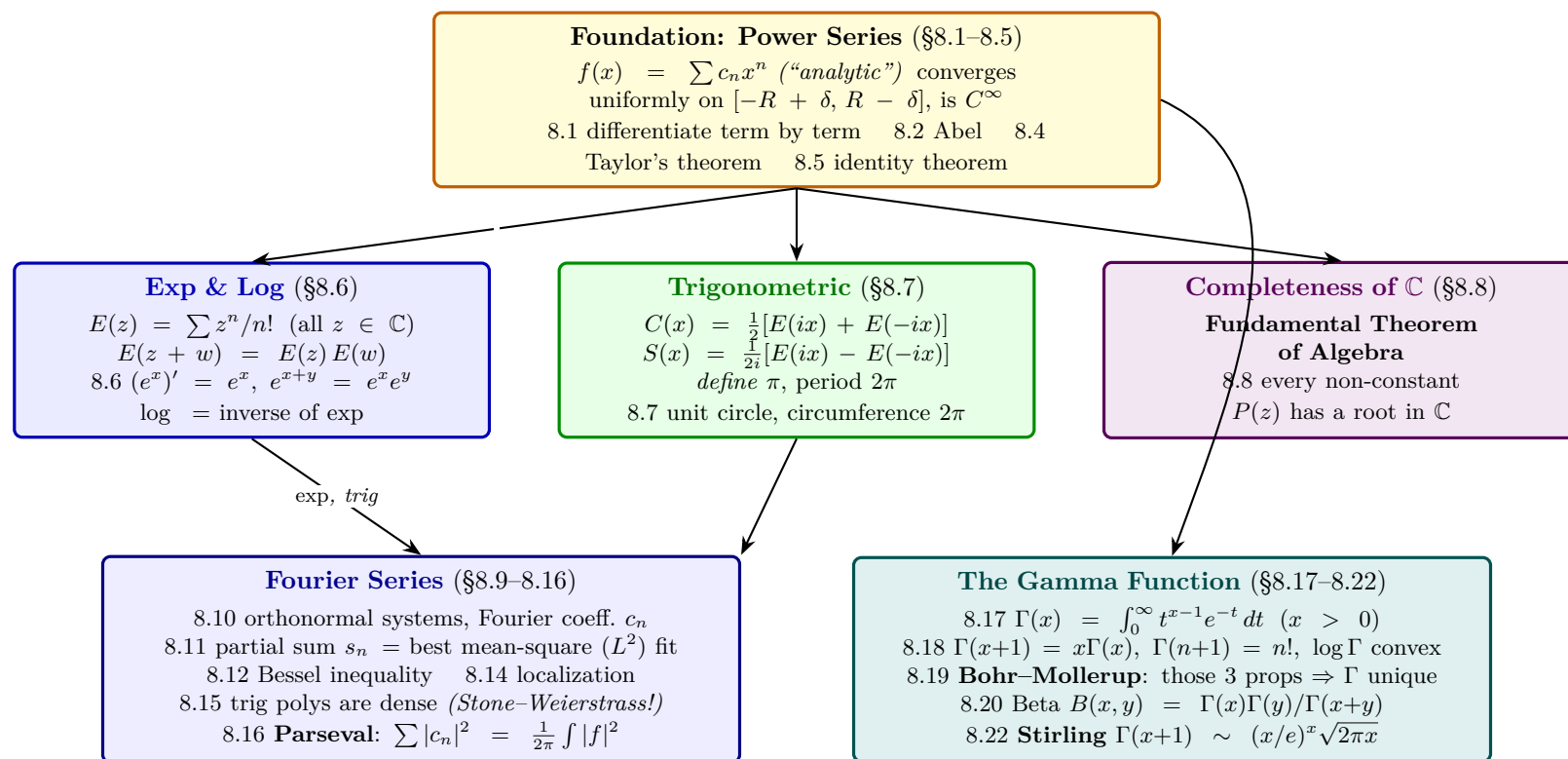
A High-Level Overview

Abstract

This is the *payoff* chapter. Using the machinery of Chapter 7 (uniform convergence, the M -test, term-by-term differentiation, the Stone–Weierstrass theorem), Rudin constructs and studies the most important functions of analysis — e^x , $\log x$, $\sin x$, $\cos x$, the number π , and the gamma function Γ — entirely from **power series**, *without* appealing to geometry. Along the way come Taylor’s theorem, the Fundamental Theorem of Algebra, Fourier series with Parseval’s theorem, and Stirling’s formula.

Visual Overview — the logical skeleton of Chapter 8

Build exp, log, sin, cos, π , Γ rigorously from series — no geometry assumed



Book pp. 172–203 • uses the Chapter 7 engine: uniform convergence, M -test, Stone–Weierstrass.

1 Power Series — the Foundation (§8.1–8.5)

Definition 1.1 (Analytic function). A function represented by a power series

$$f(x) = \sum_{n=0}^{\infty} c_n x^n, \quad \text{or more generally} \quad f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n,$$

is called *analytic*. If the first series converges on $(-R, R)$ we say f is expanded about 0; here R is the radius (interval) of convergence.

Theorem 1.2 (Term-by-term differentiation, 8.1). *If $f(x) = \sum c_n x^n$ converges for $|x| < R$, then the series converges uniformly on $[-R+\delta, R-\delta]$ for every $\delta > 0$; f is continuous and differentiable on $(-R, R)$, and*

$$f'(x) = \sum_{n=1}^{\infty} n c_n x^{n-1}.$$

Corollary 1.3. *f has derivatives of all orders on $(-R, R)$, given by repeated term-by-term differentiation, and $c_n = f^{(n)}(0)/n!$.*

Theorem 1.4 (Abel, 8.2). *If $\sum c_n$ converges and $f(x) = \sum c_n x^n$ for $|x| < 1$, then $\lim_{x \rightarrow 1^-} f(x) = \sum_{n=0}^{\infty} c_n$.*

Theorem 1.5 (Taylor's theorem, 8.4). *If $f(x) = \sum c_n x^n$ converges on $|x| < R$ and $-R < a < R$, then f can be re-expanded about a :*

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n \quad (|x-a| < R-|a|).$$

Theorem 1.6 (Identity theorem, 8.5). *If $\sum a_n x^n = \sum b_n x^n$ on a set with a limit point in $(-R, R)$, then $a_n = b_n$ for all n . (A power series is determined by its values near any point.)*

2 The Exponential and Logarithmic Functions (§8.6)

Definition 2.1 (The exponential). For every complex z ,

$$E(z) = \sum_{n=0}^{\infty} \frac{z^n}{n!} \quad (\text{converges for all } z \in \mathbb{C} \text{ by the ratio test}).$$

Multiplying absolutely convergent series gives the **addition formula**

$$E(z) E(w) = \sum_{n=0}^{\infty} \frac{(z+w)^n}{n!} = E(z+w).$$

Theorem 2.2 (Properties of e^x , 8.6). *With $e^x := E(x)$ on $\mathbb{R}$:*

- (a) e^x is continuous and differentiable, with $(e^x)' = e^x$;
- (b) e^x is strictly increasing and $e^x > 0$;
- (c) $e^{x+y} = e^x e^y$;

- (d) $e^x \rightarrow +\infty$ as $x \rightarrow +\infty$, $e^x \rightarrow 0$ as $x \rightarrow -\infty$;
 (e) $\lim_{x \rightarrow +\infty} x^n e^{-x} = 0$ for every n (e^x beats every power).

Since E is strictly increasing and differentiable, it has an inverse $L = \log$, also strictly increasing and differentiable, defined on $(0, \infty)$.

3 The Trigonometric Functions (§8.7)

Definition 3.1 (Cosine and sine, from E).

$$C(x) = \frac{1}{2}[E(ix) + E(-ix)], \quad S(x) = \frac{1}{2i}[E(ix) - E(-ix)],$$

so that $E(ix) = C(x) + iS(x)$, $|E(ix)| = 1$, $C' = -S$, $S' = C$, and $C^2 + S^2 = 1$. These coincide with $\cos x$ and $\sin x$ — but defined analytically, with no geometry.

Theorem 3.2 (Periodicity and π , 8.7). *Define π so that $\pi/2$ is the smallest positive zero of C . Then*

- (a) E is periodic with period $2\pi i$;
 (b) C and S are periodic with period 2π ;
 (c) if $0 < t < 2\pi$ then $E(it) \neq 1$;
 (d) every $z \in \mathbb{C}$ with $|z| = 1$ equals $E(it)$ for a unique $t \in [0, 2\pi)$.

Consequently $\gamma(t) = E(it)$, $0 \leq t \leq 2\pi$, is a simple closed curve tracing the unit circle, of length $\int_0^{2\pi} |\gamma'(t)| dt = 2\pi$ — giving π its usual geometric meaning.

4 Algebraic Completeness of $\mathbb{C}$ (§8.8)

Theorem 4.1 (Fundamental Theorem of Algebra, 8.8). *If $a_0, \dots, a_n \in \mathbb{C}$ with $n \geq 1$ and $a_n \neq 0$, and $P(z) = \sum_{k=0}^n a_k z^k$, then $P(z) = 0$ for some $z \in \mathbb{C}$.*

Idea of proof. Let $\mu = \inf_z |P(z)|$. Since $|P(z)| \rightarrow \infty$ as $|z| \rightarrow \infty$, the infimum is attained at some z_0 . If $P(z_0) \neq 0$ one can perturb z_0 to decrease $|P|$ (using $E(it)$ to choose a direction), contradicting minimality. Hence $P(z_0) = 0$: every non-constant complex polynomial has a root, so $\mathbb{C}$ is algebraically closed.

5 Fourier Series (§8.9–8.16)

Definition 5.1 (Trigonometric polynomial / orthonormal system, 8.9–8.10). *A trigonometric polynomial is $f(x) = a_0 + \sum_{n=1}^N (a_n \cos nx + b_n \sin nx)$. A sequence $\{\phi_n\}$ is orthonormal on $[a, b]$ if $\int_a^b \phi_n \overline{\phi_m} dx = \delta_{nm}$; e.g. $\{(2\pi)^{-1/2} e^{inx}\}$ on $[-\pi, \pi]$. The Fourier coefficients of f are $c_n = \int_a^b f \overline{\phi_n} dx$.*

Theorem 5.2 (Best mean-square approximation, 8.11). *Among all $t_n = \sum_{m=1}^n \gamma_m \phi_m$, the partial Fourier sum $s_n = \sum_{m=1}^n c_m \phi_m$ minimizes $\int_a^b |f - t_n|^2 dx$; minimum is attained iff $\gamma_m = c_m$. (Fourier partial sums are the orthogonal projection onto the span of $\phi_1, \dots, \phi_n$.)*

Theorem 5.3 (Bessel inequality (8.12) & localization (8.14)). $\sum_{n=1}^{\infty} |c_n|^2 \leq \int_a^b |f|^2 dx$, so $c_n \rightarrow 0$.
 Moreover if f satisfies a Lipschitz-type bound near x ($|f(x+t) - f(x)| \leq M|t|$), then $s_N(f; x) \rightarrow f(x)$: convergence at a point depends only on the local behaviour of f .

Theorem 5.4 (Density (8.15) & Parseval (8.16)). If f is continuous with period 2π , then for every $\varepsilon > 0$ there is a trigonometric polynomial P with $|f(x) - P(x)| < \varepsilon$ for all x (a corollary of Stone–Weierstrass). Consequently, for Riemann-integrable 2π -periodic $f \sim \sum c_n e^{inx}$,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} |f(x)|^2 dx = \sum_{n=-\infty}^{\infty} |c_n|^2 \quad (\text{Parseval}).$$

6 The Gamma Function (§8.17–8.22)

Definition 6.1 (Gamma, 8.17). For $0 < x < \infty$,

$$\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt \quad (\text{the integral converges}).$$

Theorem 6.2 (Functional equation, 8.18). On $(0, \infty)$: **(a)** $\Gamma(x+1) = x\Gamma(x)$ (integration by parts); **(b)** $\Gamma(n+1) = n!$ for $n \in \mathbb{N}$ (so Γ interpolates the factorial); **(c)** $\log \Gamma$ is convex.

Theorem 6.3 (Bohr–Mollerup characterization, 8.19). If $f > 0$ on $(0, \infty)$ satisfies $f(x+1) = xf(x)$, $f(1) = 1$, and $\log f$ is convex, then $f = \Gamma$. These three properties pin Γ down uniquely; a by-product is $\Gamma(x) = \lim_{n \rightarrow \infty} \frac{n! n^x}{x(x+1) \cdots (x+n)}$.

Theorem 6.4 (Beta function (8.20) & Stirling (8.22)). For $x, y > 0$,

$$B(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} dt = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}.$$

And for large x (hence large $n!$), **Stirling’s formula**:

$$\lim_{x \rightarrow \infty} \frac{\Gamma(x+1)}{(x/e)^x \sqrt{2\pi x}} = 1.$$

The big picture

Chapter 8 *cashes in* Chapter 7. Power series (made legitimate by uniform convergence) define $\exp$; from $\exp$ come $\log$, $\sin$, $\cos$, and π ; the same circle group $E(it)$ proves the Fundamental Theorem of Algebra. Two deeper applications follow: Fourier series (where Stone–Weierstrass guarantees trigonometric approximation and Parseval gives an L^2 isometry) and the gamma function (the factorial extended to all real — and complex — arguments, characterized by Bohr–Mollerup and estimated by Stirling).